Cyber Security

DHCP Spoofing

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https://techgirls.ece.vt.edu/slides/introduction_to_cybersecurity.html#39

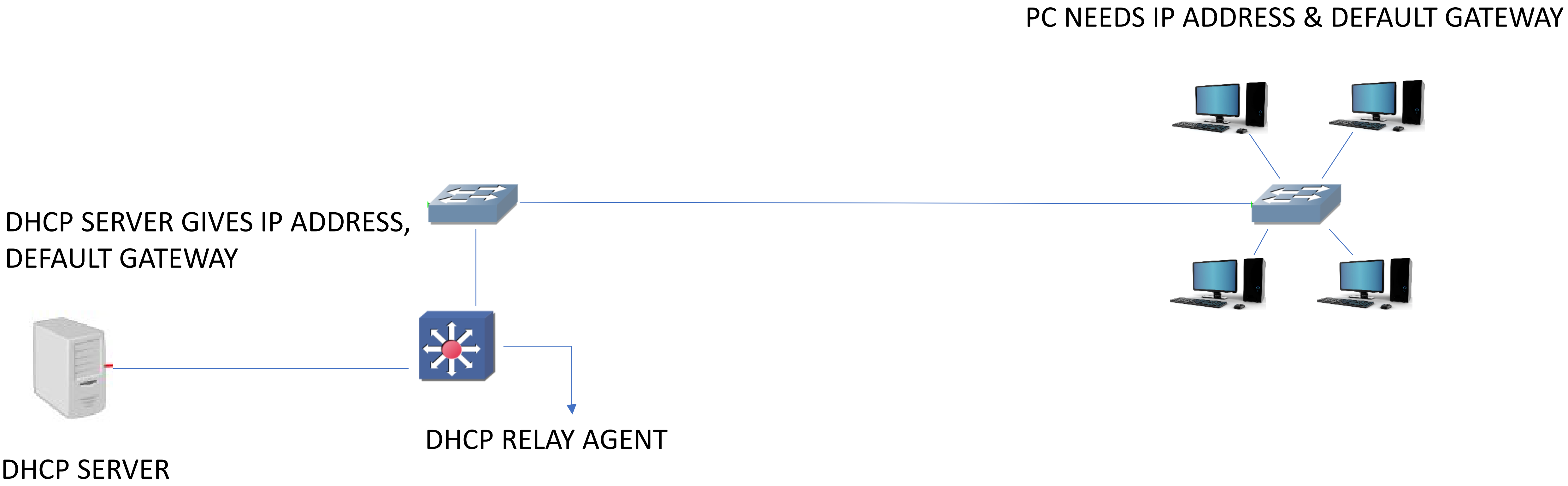
DHCP SPOOFING

-A layer two attack scenario

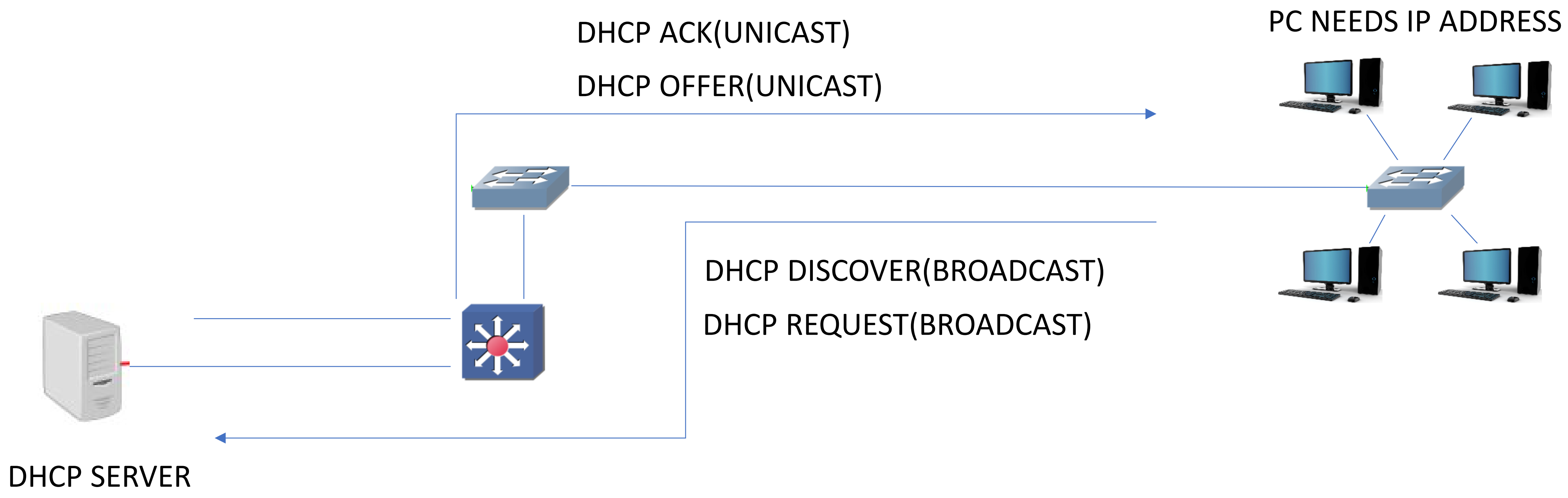
Course Material

- How DHCP works?
- How DHCP spoofing works ?
- How to prevent DHCP spoofing?

WHY DHCP ?


PC NEEDS IP ADDRESS & DEFAULT GATEWAY
DHCP SERVER GIVES IP ADDRESS, DEFAULT GATEWAY
DHCP RELAY AGENT
DHCP SERVER

DHCP OPERATION-(DORA)

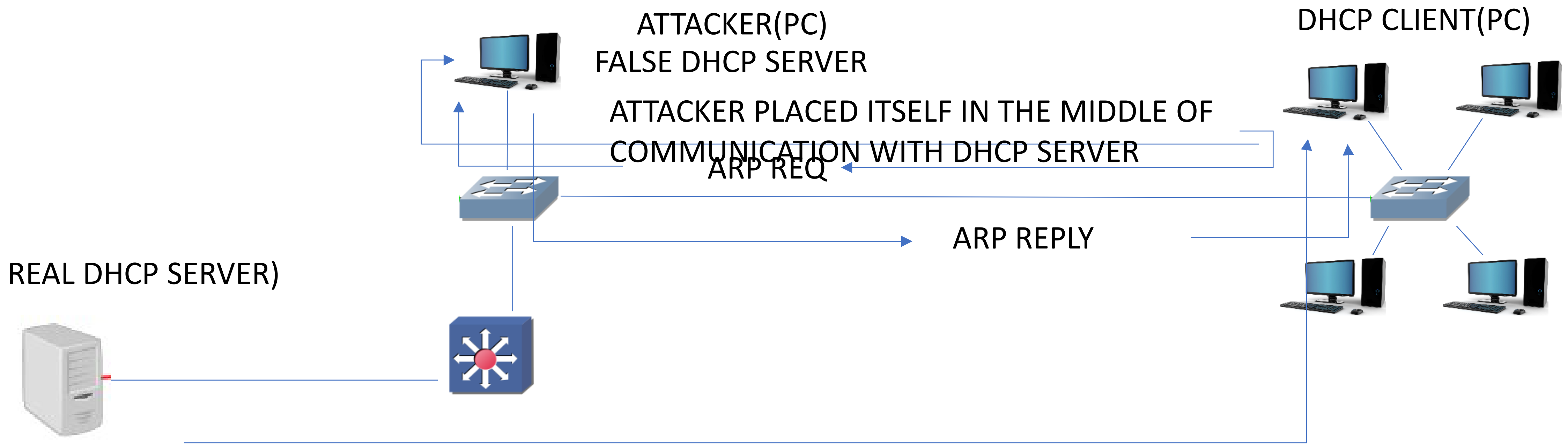


DHCP MESSEGES

Reference	Message	Use
0x01	DHCPDISCOVER	The client is looking for available DHCP servers.
0x02	DHCPOFFER	The server response to the client DHCPDISCOVER.
0x03	DHCPREQUEST	The client broadcasts to the server, requesting offered parameters from one server specifically, as defined in the packet.
0x04	DHCPDECLINE	The client-to-server communication, indicating that the network address is already in use.
0x05	DHCPACK	The server-to-client communication with configuration parameters, including committed network address.
0x06	DHCPNAK	The server-to-client communication, refusing the request for configuration parameter.
0x07	DHCPRELEASE	The client-to-server communication, relinquishing network address and canceling remaining lease.
0x08	DHCPINFORM	The client-to-server communication, asking for only local configuration parameters that the client already has externally configured as an address.

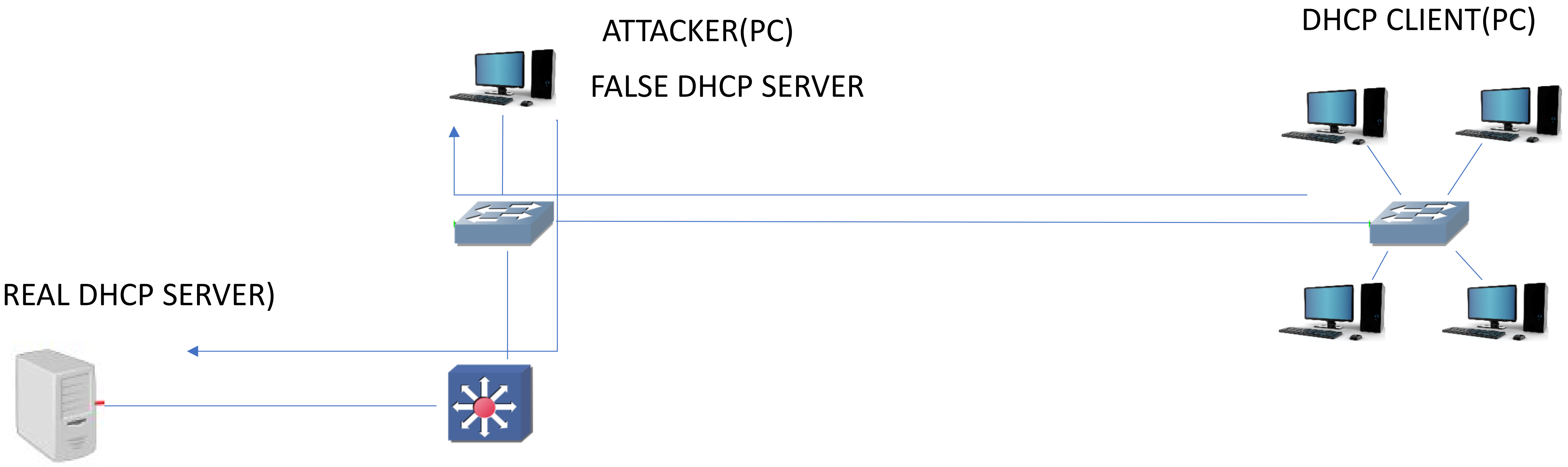
DHCP SPOOFING

I WANT IP ADDRESS,DEFUALT GATEWAY



REAL DHCP SERVER MAKES REQUEST
BUT ITS TOO LATE ATTACKER IS CLOSER THAN DHCP SERVER

DHCP SPOOFING- MAN IN THE MIDDLE ATTACK



ALL THE COMMUNICATION GOES THROUGH ATTACKER PC

DHCP SPOOFING PREVENTION

- Design network properly and enable physical security of switches
- Enable dhcp snooping.

DHCP SNOOPING(SNOOPING MEANS INVESTIGATION)

- Validates DHCP messages received from untrusted sources and filters out invalid messages.
- Rate-limits DHCP traffic from trusted and untrusted sources.
- Builds and maintains the DHCP snooping binding database, which contains information about untrusted hosts with leased IP addresses.
- Utilizes the DHCP snooping binding database to validate subsequent requests from untrusted hosts.

WHAT ARE TRUSTED AND UNTRUSTED SOURCES ?

- In an enterprise network, devices under your administrative control are trusted sources.
- Any device beyond the firewall or outside your network is an untrusted source. Host ports and unknown DHCP servers are generally treated as untrusted sources.
- The default trust state of all interfaces is untrusted. You must configure DHCP server interfaces as trusted.
- A DHCP server that is on your network without your knowledge on an untrusted port is called a *spurious DHCP server(PC with DHCP software)*
- You can detect spurious DHCP servers by sending dummy DHCPDISCOVER packets .

PACKET VALIDATION AND DROPPING PACKETS

- The switch receives DHCPOFFER, DHCPACK, DHCPNAK, or DHCPLEASEQUERY packet from a DHCP server outside the network or firewall.
- The source MAC address and the DHCP client hardware address do not match(performed only if the DHCP snooping MAC address verification option is turned on)
- DHCPRELEASE or DHCPDECLINE message from an untrusted host with an entry in the DHCP snooping binding table, and the interface information in the binding table does not match the interface on which the message was received.
- The switch receives a DHCP packet that includes a relay agent IP address that is not 0.0.0.0.

Thank You